Project: Worked examples
Title: Member checks (EN 1993-1-1)
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1. Scope and method

Resistance checks of a steel member of class 1, 2 or 3 cross-section: tension (6.2.3), compression (6.2.4) and flexural buckling about y and z (6.3.1), bending about y and z (6.2.5), shear with the high-shear reduction (6.2.6, 6.2.8), the linear cross-section interaction (6.2.1(7)) and a simplified member interaction (6.3.3 with user-entered k factors). Lateral-torsional buckling and class 4 sections are outside this sheet.

WHAT IT CALCULATES	RELATION
Axial plastic resistance	$N_{pl,Rd} = A\, f_y/\gamma_{M0}$
Bending resistance, plastic modulus for class 1–2 and elastic for class 3	$M_{c,Rd} = W\, f_y/\gamma_{M0}$
Shear resistance	$V_{pl,Rd} = A_v\, f_y/(\sqrt{3}\,\gamma_{M0})$
Bending resistance reduced by high shear, when $V_{Ed} > 0.5\, V_{pl,Rd}$	$\rho = (2V_{Ed}/V_{pl,Rd} - 1)^2$. Class 1–2 I-section with h, t_w, t_f entered: $M_{y,V,Rd} = (W_{pl,y} - \rho\, A_w^2/(4\, t_w))\, f_y/\gamma_{M0}$, $A_w = (h - 2t_f)\, t_w$, and $M_{z,c,Rd}$ is not reduced; otherwise the whole resistance × $(1 - \rho)$
Elastic critical load and non-dimensional slenderness	$N_{cr} = \pi^2 EI/L_{cr}^2$; $\bar{\lambda} = \sqrt{(A\, f_y/N_{cr})}$
Buckling reduction factor and buckling resistance	$\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$; $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$; $N_{b,Rd} = \chi\, A\, f_y/\gamma_{M1}$
Cross-section check, linear interaction	$N_{Ed}/N_{Rd} + M_{y,Ed}/M_{y,Rd} + M_{z,Ed}/M_{z,Rd} \le 1$
Member check, axial force and bending with buckling (6.61, 6.62)	$N_{Ed}/N_{b,Rd,min} + k_{yy}\, M_{y,Ed}/(W_y\, f_y/\gamma_{M1}) + k_{zz}\, M_{z,Ed}/(W_z\, f_y/\gamma_{M1}) \le 1$, the characteristic bending resistance M_{Rk} over γ_{M1} without the shear reduction

Basis: Tension[1](6.2.3); Compression[1](6.2.4); Bending moment[1](6.2.5); Shear[1](6.2.6); Bending and shear[1](6.2.8); Linear summation of utilisations[1](6.2.1(7)); Uniform members in compression: buckling resistance, buckling curves[1](6.3.1.1, 6.3.1.2); Uniform members in bending and axial compression[1](6.3.3); Imperfection factors for the buckling curves[2](§6.3.1).

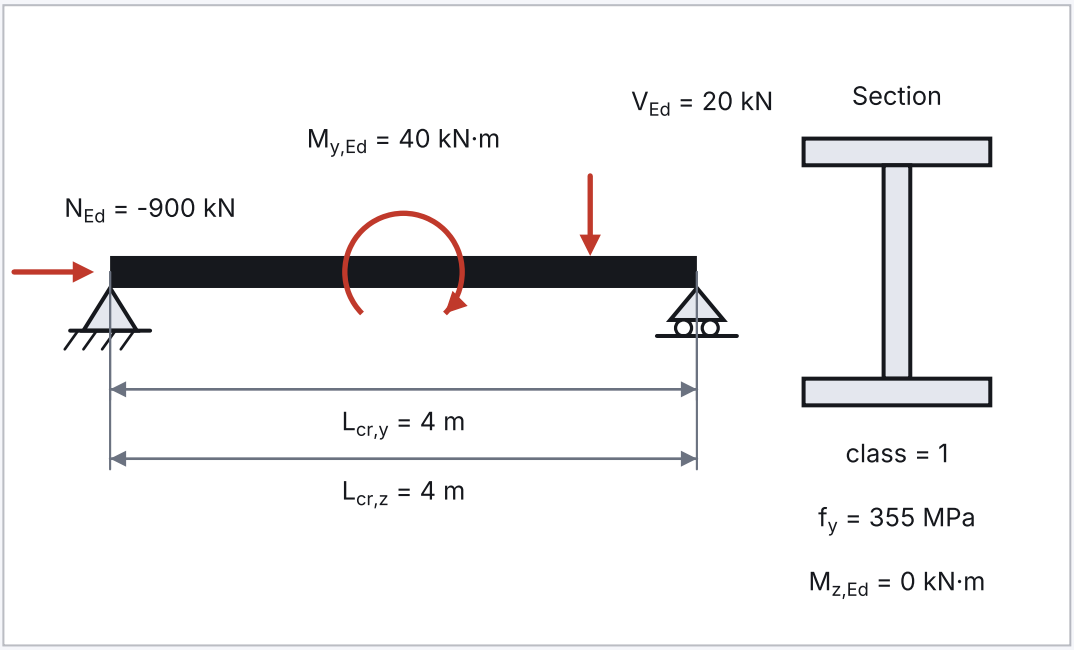


Figure 1 — A steel member under axial force, bending and shear, its buckling lengths about both axes, and its section. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
N_{Ed}	Design axial force, TENSION positive and compression negative (the opposite sign to S6's N_{Ed}, which is compression positive; S2's N_{e1} is tension positive)	-900	kN	reference: S2 frame sheet, column 1, ULS	-100000 … 100000 kN
$M_{y,Ed}$	Design moment about y-y (major)	40	kN·m	reference: S2 frame sheet, column 1, ULS	0 … 100000 kN·m
$M_{z,Ed}$	Design moment about z-z (minor)	0	kN·m	reference: S2 frame sheet	0 … 100000 kN·m
V_{Ed}	Design shear force	20	kN	reference: S2 frame sheet	0 … 100000 kN
f_y	Yield strength (for the thickness)	355	MPa	spec: BS EN 10025-2 S355, t ≤ 16 mm	200 … 700 MPa
E	Modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 3.2.6	190 … 220 GPa
γ_{M0}	Partial factor γ_{M0} (cross-section resistance)	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25
γ_{M1}	Partial factor γ_{M1} (member buckling)	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25
$section_{class}$	Cross-section class	1		assumption: UC 203×203×46 S355 is class 1 in compression (Blue Book).	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
A	Area A	58.7	cm^2	reference: S1 section sheet / Blue Book UC 203×203×46	1 … 100000 cm^2
$W_{pl,y}$	$W_{pl,y}$	497	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{el,y}$	$W_{el,y}$	450	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{pl,z}$	$W_{pl,z}$	231	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{el,z}$	$W_{el,z}$	152	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
I_y	I_y	4568	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
I_z	I_z	1548	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
h	Depth h (shear-area helper; leave blank when A_v is given)	203.2	mm	spec: BS EN 10365 UC 203×203×46	1 … 5000 mm
b	Width b (helper)	203.6	mm	spec: BS EN 10365 UC 203×203×46	1 … 5000 mm
t_w	Web thickness (helper)	7.2	mm	spec: BS EN 10365 UC 203×203×46	0.1 … 500 mm
t_f	Flange thickness (helper)	11	mm	spec: BS EN 10365 UC 203×203×46	0.1 … 500 mm
r	Root radius (helper)	10.2	mm	spec: BS EN 10365 UC 203×203×46	0 … 500 mm
η	Shear-area factor η	1		spec: UK NA to BS EN 1993-1-5 NA.2.4	1 … 1.2
$L_{cr,y}$	Buckling length about y-y	4	m	assumption: Pinned both ends: L_cr = L = 4 m.	0.1 … 100 m
$L_{cr,z}$	Buckling length about z-z	4	m	assumption: Pinned both ends: L_cr = L = 4 m.	0.1 … 100 m
$curve_y$	Buckling curve about y-y	b		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b ≤ 1.2, t_f ≤ 100, S355, y-y	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$curve_z$	Buckling curve about z-z	c		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b ≤ 1.2, t_f ≤ 100, S355, z-z	any
k_{yy}	Interaction factor k_{yy} (member check)	1		assumption: Simplified interaction: k_yy = 1.0.	0.5 … 2
k_{zz}	Interaction factor k_{zz} (member check)	1		assumption: Simplified interaction: k_zz = 1.0.	0.5 … 2
ltb	Compression flange against lateral-torsional buckling (not restrained: outside this sheet)	Restrained		default	any
γ_{M2}	Partial factor γ_{M2} (net-section fracture)	1.1		default	1 … 1.35

3. Assumptions

A1. The cross-section class is as stated by the user (1, 2 or 3); class 4 effective properties are not derived here. (6.2.2.5 effective widths are outside this sheet.)
A2. The compression flange is restrained against lateral-torsional buckling, as the input ltb says. (6.3.2 is outside this sheet: a member that is not restrained is checked on the LTB interaction sheet.)
A3. A k left at 1.0 under $N_{Ed}/N_{b,Rd}$ above 0.3 is replaced by the Annex A/B upper bound $C_m(1 + 0.8n)$. (6.3.3 simplified: the smaller flexural-buckling reduction is used on the axial term and the factors are not derived.)
A4. Imperfection factors per curve are the textbook values and may be overridden; partial factors default to 1.0 (UK National Annex) and may be overridden. (User-editable with provenance.)
A5. $section_{class}$ (Cross-section class) is taken as 1. (UC 203×203×46 S355 is class 1 in compression (Blue Book).)
A6. $L_{cr,y}$ (Buckling length about y-y) is taken as 4 m. (Pinned both ends: L_{cr} = L = 4 m.)
A7. $L_{cr,z}$ (Buckling length about z-z) is taken as 4 m. (Pinned both ends: L_{cr} = L = 4 m.)
A8. k_{yy} (Interaction factor k_{yy} (member check)) is taken as 1. (Simplified interaction: k_{yy} = 1.0.)
A9. k_{zz} (Interaction factor k_{zz} (member check)) is taken as 1. (Simplified interaction: k_{zz} = 1.0.)
A10. The compression flange is restrained against lateral-torsional buckling, as the input ltb says. (Input ltb.)

4. Working

[28] $r_h = r$
$= 10.2\ mm$
$= 10.2\ mm$
Root radius used by the helper

[29] $A_{v,rolled} = A - 2 * b * t_f + (t_w + 2 * r_h) * t_f$ [1](6.2.6(3))
$= 58.7\ cm^2 - 2 \times 203.6\ mm \times 11\ mm + (7.2\ mm + 2 \times 10.2\ mm) \times 11\ mm$
$= 16.944\ cm^2$
Rolled I shear area, load parallel to the web

[30] $A_{v,min} = \eta * (h - 2 * t_f) * t_w$ [1](6.2.6(3))
$= 1 \times (203.2\ mm - 2 \times 11\ mm) \times 7.2\ mm$
$= 13.046\ cm^2$
Lower bound $\eta\ h_w\ t_w$

[31] $A_v = \max(A_{v,rolled}, A_{v,min})$
$= \max(16.944\ cm^2, 13.046\ cm^2)$
$= 16.944\ cm^2$
Shear area
[32] $N_{pl,Rd} = A * f_y / \gamma_{M0}$ [1](6.2.3(2), 6.2.4(2))
$= 58.7\ cm^2 \times 355\ MPa / 1$
$= 2083.8\ kN$
Plastic axial resistance
[33] $W_y = W_{pl,y}$ [1](6.2.5(2))
$= 497\ cm^3$
$= 497\ cm^3$
Section modulus about y (plastic, class 1–2)
[34] $W_z = W_{pl,z}$ [1](6.2.5(2))
$= 231\ cm^3$
$= 231\ cm^3$
Section modulus about z (plastic)
[35] $V_{pl,Rd} = A_v * f_y / (\sqrt{3} * \gamma_{M0})$ [1](6.2.6(2))
$= 16.944\ cm^2 \times 355\ MPa / (\sqrt{3} \times 1)$
$= 347.28\ kN$
Plastic shear resistance
[36] $u_V = V_{Ed} / V_{pl,Rd}$
$= 20\ kN / 347.28\ kN$
$= 0.05759$
Shear utilisation
[37] $\rho = 0$ [1](6.2.8(2))
$= 0$
Shear reduction factor ($V_{Ed} \le 0.5\ V_{pl,Rd}$)
[38] $M_{y,c,Rd} = (1 - \rho) * W_y * f_y / \gamma_{M0}$ [1](6.2.5(2), 6.2.8)
$= (1 - 0) \times 497\ cm^3 \times 355\ MPa / 1$
$= 176.44\ kN{\cdot}m$
Bending resistance about y
[39] $M_{z,c,Rd} = (1 - \rho) * W_z * f_y / \gamma_{M0}$ [1](6.2.5(2))
$= (1 - 0) \times 231\ cm^3 \times 355\ MPa / 1$
$= 82.005\ kN{\cdot}m$
Bending resistance about z
[40] $u_{M,y} = M_{y,Ed} / M_{y,c,Rd}$
$= 40\ kN{\cdot}m / 176.44\ kN{\cdot}m$
$= 0.22671$
Bending utilisation, y
[41] $u_{M,z} = M_{z,Ed} / M_{z,c,Rd}$
$= 0\ kN{\cdot}m / 82.005\ kN{\cdot}m$
$= 0$
Bending utilisation, z
[42] $u_N = \mathrm{abs}(N_{Ed}) / N_{pl,Rd}$
$= \mathrm{abs}(-900\ kN) / 2083.8\ kN$
$= 0.43189$
Compression (cross-section) utilisation
[43] $u_{section} = u_N + u_{M,y} + u_{M,z}$ [1](6.2.1(7))
$= 0.43189 + 0.22671 + 0$
$= 0.65861$
Linear cross-section interaction
[46] α_y = imperfection factor, curve b [1](6.3.1.2)
= Designers' Guide §6.3.1
$= 0.34$
Imperfection factor about y

[47] α_z = imperfection factor, curve c [1](6.3.1.2)
= Designers' Guide §6.3.1
= 0.49
Imperfection factor about z
[48] $N_{cr,y}$ = pi^2 * E * I_y / $L_{cr,y}$^2 [1](6.3.1.2(1))
= π^2 × 210 GPa × 4568 cm⁴ / (4 m)^2
= 5917.3 kN
Elastic critical force about y
[49] λ_y = sqrt(A * f_y / $N_{cr,y}$) [1](6.3.1.2(1))
= sqrt(58.7 cm² × 355 MPa / 5917.3 kN)
= 0.59343
Non-dimensional slenderness about y
[50] Φ_y = 0.5 * (1 + α_y * (λ_y - 0.2) + λ_y^2) [1](6.3.1.2(1))
= 0.5 × (1 + 0.34 × (0.59343 - 0.2) + 0.59343^2)
= 0.74296
Φ about y
[51] χ_y = min(1, 1 / (Φ_y + sqrt(Φ_y^2 - λ_y^2))) [1](6.3.1.2(1))
= min(1, 1 / (0.74296 + sqrt(0.74296^2 - 0.59343^2)))
= 0.84034
Reduction factor χ about y
[52] $N_{b,y,Rd}$ = χ_y * A * f_y / γ_{M1} [1](6.3.1.1(3))
= 0.84034 × 58.7 cm² × 355 MPa / 1
= 1751.1 kN
Buckling resistance about y
[53] $N_{cr,z}$ = pi^2 * E * I_z / $L_{cr,z}$^2 [1](6.3.1.2(1))
= π^2 × 210 GPa × 1548 cm⁴ / (4 m)^2
= 2005.3 kN
Elastic critical force about z
[54] λ_z = sqrt(A * f_y / $N_{cr,z}$) [1](6.3.1.2(1))
= sqrt(58.7 cm² × 355 MPa / 2005.3 kN)
= 1.0194
Non-dimensional slenderness about z
[55] Φ_z = 0.5 * (1 + α_z * (λ_z - 0.2) + λ_z^2) [1](6.3.1.2(1))
= 0.5 × (1 + 0.49 × (1.0194 - 0.2) + 1.0194^2)
= 1.2204
Φ about z
[56] χ_z = min(1, 1 / (Φ_z + sqrt(Φ_z^2 - λ_z^2))) [1](6.3.1.2(1))
= min(1, 1 / (1.2204 + sqrt(1.2204^2 - 1.0194^2)))
= 0.52876
Reduction factor χ about z
[57] $N_{b,z,Rd}$ = χ_z * A * f_y / γ_{M1} [1](6.3.1.1(3))
= 0.52876 × 58.7 cm² × 355 MPa / 1
= 1101.9 kN
Buckling resistance about z
[58] $N_{b,Rd}$ = min($N_{b,y,Rd}$, $N_{b,z,Rd}$)
= min(1751.1 kN, 1101.9 kN)
= 1101.9 kN
Governing buckling resistance
[59] $u_{buckling}$ = abs(N_{Ed}) / $N_{b,Rd}$ [1](6.3.1.1(1))
= abs(-900 kN) / 1101.9 kN
= 0.8168
Flexural buckling utilisation

[60] $M_{y,b,Rd} = W_y * f_y / \gamma_{M1}$ [1](6.3.3, Eq. 6.61)
= 497 cm³ × 355 MPa / 1
= 176.44 kN·m
Characteristic bending resistance about y over γ_{M1} (member check, no shear reduction)
[61] $M_{z,b,Rd} = W_z * f_y / \gamma_{M1}$ [1](6.3.3, Eq. 6.62)
= 231 cm³ × 355 MPa / 1
= 82.005 kN·m
Characteristic bending resistance about z over γ_{M1} (member check, no shear reduction)
[62] $u_{member} = abs(N_{Ed}) / N_{b,Rd} + k_{yy} * M_{y,Ed} / M_{y,b,Rd} + k_{zz} * M_{z,Ed} / M_{z,b,Rd}$ [1](6.3.3(4))
= abs(-900 kN) / 1101.9 kN + 1 × 40 kN·m / 176.44 kN·m + 1 × 0 kN·m / 82.005 kN·m
= 1.0435
Simplified member interaction

5. Results

SYMBOL	VALUE	DESCRIPTION
A_v	16.94 cm²	Shear area
$N_{pl,Rd}$	2083.8 kN	$N_{pl,Rd}$
$V_{pl,Rd}$	347.28 kN	$V_{pl,Rd}$
ρ	0	ρ
$M_{y,c,Rd}$	176.44 kN·m	$M_{y,c,Rd}$
$M_{z,c,Rd}$	82.01 kN·m	$M_{z,c,Rd}$
$M_{y,b,Rd}$	176.44 kN·m	$M_{y,Rk}/\gamma_{M1}$
$M_{z,b,Rd}$	82.01 kN·m	$M_{z,Rk}/\gamma_{M1}$
$N_{cr,y}$	5917.3 kN	$N_{cr,y}$
$N_{cr,z}$	2005.3 kN	$N_{cr,z}$
λ_y	0.59343	$\bar{\lambda}_y$
λ_z	1.0194	$\bar{\lambda}_z$
χ_y	0.84034	χ_y
χ_z	0.52876	χ_z
$N_{b,y,Rd}$	1751.1 kN	$N_{b,y,Rd}$
$N_{b,z,Rd}$	1101.9 kN	$N_{b,z,Rd}$
$N_{b,Rd}$	1101.9 kN	$N_{b,Rd}$
u_N	0.43	Axial utilisation
$u_{M,y}$	0.23	Bending y utilisation
$u_{M,z}$	0.00	Bending z utilisation
u_V	0.06	Shear utilisation
$u_{section}$	0.66	Cross-section interaction
$u_{buckling}$	0.82	Buckling utilisation
u_{member}	1.05	Member interaction (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Member interaction (simplified)	$u_{member} = 1.05$	$u_{member} \le 1$	1.05	[1](6.3.3)
PASS	Flexural buckling $N_{Ed} \le N_{b,Rd}$	$u_{buckling} = 0.82$	$u_{buckling} \le 1$	0.82	[1](6.3.1.1)
PASS	Cross-section interaction	$u_{section} = 0.66$	$u_{section} \le 1$	0.66	[1](6.2.1(7))
PASS	Compression $N_{Ed} \le N_{c,Rd}$	$u_N = 0.43$	$u_N \le 1$	0.43	[1](6.2.4)
PASS	Bending y $M_{y,Ed} \le M_{y,c,Rd}$	$u_{M,y} = 0.23$	$u_{M,y} \le 1$	0.23	[1](6.2.5)
PASS	Shear $V_{Ed} \le V_{pl,Rd}$	$u_V = 0.06$	$u_V \le 1$	0.06	[1](6.2.6)
PASS	Bending z $M_{z,Ed} \le M_{z,c,Rd}$	$u_{M,z} = 0.00$	$u_{M,z} \le 1$	0.00	[1](6.2.5)

7. Applicability

WARNING k_{yy}: entered as 1.0, but Annex A or B give k up to 1.411 here. With it the interaction would be 1.137 against 1.044 shown. [k-default]

Defaults used (inputs left empty):
ltb: not entered, so the default of 'Restrained' is used.
γ_{M2}: not entered, so the default of 1.1 is used.

8. Verification

```
Template struct.en1993-member@1.4.0 Content hash fdf55f0878a74ece Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-1: General Rules and Rules for Buildings. BS EN 1993-1-1. BSI.
2. Gardner L, Nethercot DA. Designers' Guide to EN 1993-1-1: Eurocode 3: Design of Steel Structures, General Rules and Rules for Buildings. Thomas Telford; 2005.